TERMS OF REFERENCE

Graduate Research Assistant - FungaFrass Bio (FF-Bio) Project

Mzuri Organics (Gare Holdings Ltd) | June 2026

Assignment Title	Part-Time Graduate Research Assistant
Project	FungaFrass Bio (FF-Bio): Scaling Fungus-Rich Biofertilizers for Safer and Resilient Informal Vegetable Value Chains
Client / Organisation	Mzuri Organics (Gare Holdings Ltd)
Project Location	Kakamega, Bungoma, Siaya, Uasin Gishu and Vihiga Counties, Kenya
Duty Station	Mzuri Organics coordination base and project field counties, with scheduled field travel and desk-based research work
Engagement Type	Part-time, project-based graduate research support
Duration / Level of Effort	Eight (8) months; ten (10) working days per month, subject to project needs, field schedules and satisfactory performance
Monthly Stipend	KES 50,000 per month for the agreed 10-day monthly input; subject to applicable tax treatment and contract terms
Field Facilitation	Approved field travel and project facilitation will be provided separately in line with FF-Bio procedures and documentation requirements
Reporting Line	Project Coordinator
Technical Supervision	M&E Lead and relevant technical specialists: Soil Scientist, Soil Microbiologist, Soil Health & IPM Expert, Crop Nutrition Specialist, and Plant Pathology & Agricultural Biometry Specialist
Preferred Level	Master’s student or recent Master’s graduate

1. Background

Mzuri Organics (Gare Holdings Ltd) is implementing the FungaFrass Bio (FF-Bio) sub-project under FS4Africa Open Call 1, funded by the European Union. FF-Bio is a multi-county field demonstration and research initiative focused on validating fungus-rich biofertilizer systems and soil-health-based Integrated Pest Management (IPM) approaches as practical pathways for reducing pesticide dependence, improving soil biological health, supporting safer vegetable production, and strengthening farmer profitability in informal vegetable value chains.

The project will be implemented across Kakamega, Bungoma, Siaya, Uasin Gishu and Vihiga Counties. It involves participatory demonstration gardens, paired treatment/control plots, farmer training, bio-input deployment, IPM coaching, soil and crop monitoring, pesticide residue testing, economic analysis, adoption assessment, stakeholder dissemination, and evidence generation for policy, scaling and scientific publication.

Mzuri Organics seeks to engage a Graduate Research Assistant to provide structured field research, monitoring, data management, sample tracking, analysis and technical documentation support. The assignment is designed to offer meaningful applied research exposure while strengthening the quality, discipline and continuity of FF-Bio evidence generation.

2. Purpose of the Assignment

The purpose of this assignment is to provide dedicated graduate-level research support to the FF-Bio project on a part-time basis. The Graduate Research Assistant will support field implementation, data collection, data quality assurance, sample tracking, evidence documentation, preliminary analysis, reporting, and development of scientific and knowledge products.

A key final ambition of FF-Bio is to develop a science-backed manuscript for submission to a peer-reviewed scientific journal. The Graduate Research Assistant will therefore contribute to literature review, dataset organisation, preliminary analysis, visualisation, preparation of tables and figures, and manuscript drafting support under the guidance of the Project Coordinator, M&E Lead and technical experts.

3. Specific Objectives of the Role

1. Support timely and high-quality implementation of FF-Bio field research and demonstration activities across assigned counties and sites.
2. Strengthen data collection, data cleaning, documentation and evidence management across project sites.

3. Support soil, crop, pesticide-use, residue, yield, economic, adoption and farmer perception data streams.
4. Assist with sample planning, labelling, chain-of-custody documentation and laboratory follow-up for soil, crop and residue testing.
5. Contribute to technical interpretation and organisation of findings for project reports, policy/practice briefs, learning products and scientific manuscript development.
6. Ensure farmer-facing research activities are implemented in line with approved ethics, informed consent, confidentiality, data protection and do-no-harm protocols.

4. Scope of Work and Key Responsibilities

4.1 Research Planning and Field Preparation

- Assist in preparing field visit schedules, farmer/site files, plot identification records, data collection tools, sample labels, monitoring checklists and field evidence folders.
- Support the M&E Lead and technical team in operationalising approved protocols, SOPs, treatment records, sampling plans and reporting templates.
- Maintain a site/farmer tracking file for assigned demonstration sites, including farmer IDs, county/site details, crop, treatment type, visit history and key follow-up actions.

4.2 Field Research and Demonstration Support

- Support establishment, monitoring and documentation of FF-Bio demonstration plots across assigned counties and sites.
- Assist with verification of plot layout, treatment allocation, treatment/control compliance, crop management records and field schedules.
- Support farmer training, farmer follow-up visits, field days and site-level learning activities as assigned by the Project Coordinator and technical leads.
- Conduct or support farm observations covering crop condition, pest/disease status, input storage, treatment adherence, irrigation, labour use and plot management.

4.3 Data Collection, Data Quality and Documentation

- Support baseline, midline and endline data collection using approved digital or paper tools.
- Assist in capturing farmer demographic data, farm practices, fertilizer use, pesticide use, spray frequency, active ingredients where available, labour, irrigation, yields, market prices, adoption metrics, gross margins and farmer perceptions.
- Review field data for completeness, consistency, accuracy and logical errors; flag and help correct data issues promptly.
- Maintain organised digital and physical evidence folders, including field notes, attendance sheets, photos where consent is obtained, GPS records, sample logs and reports.

4.4 Soil, Crop and Residue Sampling Support

- Support soil and crop sample collection in line with approved project protocols, including correct sampling timing, depth, plot, crop stage, treatment category, labelling and packaging.
- Assist with sample tracking for soil physicochemical analysis, soil biological indicators, crop observations and pesticide residue testing.
- Ensure sample records capture required metadata, including date, county, site, farmer ID, treatment, crop, sample type, GPS/location where applicable, and relevant field observations.
- Maintain sample chain-of-custody records and coordinate with the Project Coordinator, Soil Scientist, Soil Microbiologist and laboratory focal persons.

4.5 Data Analysis and Research Support

- Assist with data cleaning, coding, summary tables, visualisations and preliminary descriptive analysis.
- Support preparation of tables and charts on crop performance, pesticide-use reduction, residue compliance, soil health indicators, farmer adoption and economic outcomes.
- Where capacity allows, support statistical analysis using Excel, R, Python, SPSS, STATA, Kobo/ODK exports or other approved tools under supervision of the technical team.
- Prepare case notes, field observations and concise technical memos that help translate raw field evidence into project learning.

4.6 Scientific Publication and Knowledge Dissemination

- Conduct a targeted and well-referenced literature review on fungus-rich biofertilizers, soil biological health, IPM, pesticide reduction, MRL compliance, food safety, informal vegetable markets and farmer adoption.

- Develop an annotated literature review matrix and reference library to support manuscript preparation.
- Contribute to preparation of a science-backed manuscript for submission to a peer-reviewed journal, including drafting background sections, methods summaries, result summaries, tables, figures and references under supervision.
- Support preparation of project learning briefs, technical notes, conference abstracts, posters, policy/practice briefs and evidence summaries where required.
- Apply responsible scientific writing practices, including plagiarism avoidance, proper citation, data integrity, responsible authorship and acknowledgement of contributors.

4.7 Reporting, Coordination and Timesheets

- Prepare concise weekly updates where required, monthly reports, field visit notes and activity summaries for the Project Coordinator and M&E Lead.
- Participate in project review meetings and provide updates on data, field progress, sample tracking, emerging observations and research outputs.
- Submit a monthly timesheet, work summary and deliverable evidence for the agreed 10 working days per month before stipend processing.
- Support documentation of farmer trainings, field days and stakeholder demonstrations through attendance records, photos where consent is obtained, proceedings notes and field evidence.

4.8 Ethics, Consent and Safeguarding

- Ensure all farmer-facing activities follow approved informed consent and data protection protocols.
- Respect farmer privacy, avoid coercion, avoid leading questions and avoid making financial, market, yield or employment promises to participants.
- Use farmer IDs and anonymised datasets as instructed; never share raw identifiable farmer data outside authorised project channels.
- Report field incidents, farmer complaints, consent issues, data risks or safeguarding concerns immediately to the Project Coordinator or M&E Lead.

5. Expected Deliverables and Indicative Timeline

Timeline	Deliverable	Expected Output
Month 1	Inception support and personal workplan	Agreed personal workplan for 10 days/month, field calendar, orientation completed, tools reviewed, data/sample tracking templates prepared, and literature review plan agreed.
Months 1-2	Baseline support package	Baseline field notes, uploaded/cleaned forms, attendance/evidence folders, plot verification notes, site/farmer profile summary and sample logs where applicable.
Months 1-3	Annotated literature review matrix	Structured literature review and reference library covering biofertilizers, soil biology, IPM, pesticide reduction, food safety, residue compliance and vegetable productivity.
Monthly	Monthly activity report and timesheet	Monthly summary of the 10-day input, sites visited, tasks completed, datasets handled, deliverables produced, challenges, risks and next steps.
As scheduled	Sampling and monitoring support records	Complete soil/crop sample logs, labels, GPS/site metadata, chain-of-custody records, spray logs, input/cost records and crop monitoring records for assigned events.
Midline period	Midline support outputs	Midline data collection support, cleaned records, preliminary observations on crop performance, pesticide-use trends, IPM adoption and farmer feedback.
Months 4-7	Preliminary analysis products	Clean datasets, summary tables, charts and concise technical summaries for review by the M&E Lead and technical team.
Months 6-8	Scientific manuscript support package	Draft manuscript outline, literature review text, methods summary, draft tables/figures, reference list and results summary for project review.
Month 8	Final graduate assistant report	Concise final technical report summarising field implementation support, datasets handled, evidence products, manuscript support, lessons, limitations and recommendations.
Month 8	Knowledge dissemination contribution	At least one contribution to a technical brief, conference abstract, poster, policy/practice brief, stakeholder evidence note or learning product.

6. Level of Effort and Work Arrangement

- This is a part-time assignment expected to run for eight (8) months with a ten (10) working day input per month, subject to project workplans, field calendars and satisfactory performance.

- The 10 days per month may include fieldwork, desk-based data management, literature review, technical meetings, report preparation and manuscript support, as agreed with the Project Coordinator.
- The Graduate Research Assistant may be required to travel across project counties, subject to approved field schedules and facilitation arrangements.
- A monthly timesheet, work summary and deliverables log shall be submitted before stipend processing.
- The engagement does not create an automatic employment relationship and does not guarantee continuation beyond the project period.

7. Stipend and Facilitation

The Graduate Research Assistant will receive a monthly stipend of KES 50,000 for the approved 10-day monthly input during the engagement period, subject to satisfactory performance, submission of required reports/timesheets and applicable tax or statutory treatment under the signed contract.

Approved field facilitation for travel, meals, communication and other project-related costs will be handled separately in line with FF-Bio procedures and must be supported by appropriate documentation where required. The stipend is not intended to cover approved project field costs that have been separately authorised.

8. Supervision and Reporting Arrangements

- The Graduate Research Assistant will report administratively to the Project Coordinator.
- Technical guidance will be provided by the M&E Lead and relevant specialists, including the Soil Scientist, Soil Microbiologist, Soil Health & IPM Expert, Crop Nutrition Specialist, and Plant Pathology & Agricultural Biometry Specialist.
- The Assistant will participate in scheduled team briefings, field planning meetings, data review meetings and technical debriefs as required.
- All project communication with farmers, laboratories, partners, county officials or external actors must be coordinated through the Project Coordinator or authorised project lead.
- All field schedules, data collection plans, sample tracking, analysis outputs and manuscript contributions shall be reviewed through the approved project reporting structure.

9. Scientific Manuscript, Publication and Authorship Guidance

- One of the expected final outputs of FF-Bio is a science-backed manuscript for submission to a peer-reviewed scientific journal.
- The Graduate Research Assistant is expected to support manuscript preparation through literature review, data organisation, preliminary analysis, tables, figures, references, technical summaries and draft text as assigned.
- Journal submission and acceptance are outside the sole control of the Graduate Research Assistant and therefore cannot be guaranteed as an individual deliverable.
- Authorship shall be determined based on substantive intellectual contribution, data contribution, analysis, writing contribution, project ownership, technical leadership and final approval by Mzuri Organics and the technical team.
- The Graduate Research Assistant may be considered for authorship or acknowledgement where their contribution meets the agreed publication and authorship standards.
- No project data, manuscript draft, abstract, poster, presentation or publication shall be submitted externally without written approval from Mzuri Organics and the designated project leadership.
- All publications and dissemination outputs must respect project confidentiality, farmer anonymisation, consent conditions, funder acknowledgement requirements and applicable journal ethics standards.

10. Required Qualifications and Experience

- Ongoing or recently completed Master's degree in Soil Science, Agronomy, Horticulture, Crop Protection, Environmental Science, Agricultural Extension, Agricultural Biotechnology, Applied Microbiology, Food Safety, Plant Pathology, Agricultural Economics or a closely related discipline.
- Strong interest in soil health, regenerative agriculture, food safety, bio-inputs, IPM, pesticide reduction, vegetable production, smallholder agriculture and informal vegetable value chains.
- Experience or demonstrated ability in field research, farmer engagement, data collection, data cleaning, literature review, scientific writing or technical report writing.
- Good working knowledge of Microsoft Excel; familiarity with KoboToolbox, ODK, Google Forms, R, Python, GIS, SPSS, STATA or statistical software will be an added advantage.

- Ability to work with farmer groups respectfully, communicate clearly and travel to rural field sites across Western Kenya.
- Strong organisational skills, attention to detail, reliability, integrity and ability to meet deadlines under field conditions.
- Good writing skills in English; ability to communicate in Kiswahili and relevant local languages used in the project counties will be an advantage.

11. Added Advantage

- Evidence of academic publishing experience or contribution to a journal article, manuscript draft, thesis publication, conference paper, poster, technical research paper or policy/practice brief.
- Experience with soil sampling, crop monitoring, pest/disease scouting, pesticide-use data collection, residue studies, field trials, RCBD or paired plot designs, or farmer participatory research.
- Basic ability to analyse and visualise agronomic, economic, biological or survey data.
- Familiarity with reference management tools such as Zotero, Mendeley, EndNote or similar platforms.
- Interest in developing academic outputs, thesis work, posters or manuscripts from approved anonymised project data under Mzuri Organics publication rules.

12. Ethics, Confidentiality and Data Protection

- Participation in farmer interviews, FGDs, sampling, photos or recordings must be voluntary and based on informed consent.
- The Assistant must use approved data collection tools only and must not alter protocols without authorisation.
- Farmer names and identifiers must not appear in analysis outputs or reports unless explicitly approved and consented.
- All data must be stored in secure, password-protected systems and shared only with authorised project personnel.
- The Assistant must not promise financial compensation, market access, yield improvements, employment or other benefits to farmers beyond what has been formally approved.
- Any incident, farmer complaint, consent concern, data breach or field safety issue must be reported immediately to the Project Coordinator or M&E Lead.
- Academic use of project data requires prior written approval from Mzuri Organics and must comply with anonymisation, ethics, confidentiality, authorship and acknowledgement requirements.
- All research outputs, datasets, photos, reports, manuscript drafts and related intellectual outputs prepared under this assignment shall remain subject to Mzuri Organics' project ownership, approval and publication rules.

13. Performance Indicators

Performance Area	Assessment Focus
Data quality	Completeness, consistency and accuracy of submitted field records, sample logs and datasets.
Timeliness	Submission of monthly updates, timesheets, sample logs and assigned outputs within agreed deadlines.
Protocol adherence	Correct use of approved tools, SOPs, farmer IDs, consent procedures and sample tracking systems.
Field reliability	Attendance, preparedness, respectful farmer engagement and responsiveness to field schedules.
Analytical contribution	Quality of summaries, tables, charts, field observations, literature review matrix and learning notes.
Scientific writing contribution	Quality of referenced literature review, draft text, tables, figures and manuscript support outputs.
Ethics compliance	No avoidable breach of confidentiality, consent, safety, data protection or publication approval standards.

14. Application Requirements

- Cover letter or motivation letter describing suitability, interest in FF-Bio and relevant research/field experience.
- Updated CV, not more than three pages, including academic background, field research experience, software/data skills and publications or writing outputs where applicable.
- Academic transcript or proof of enrolment/completion at Master's level.
- One writing sample, preferably a thesis chapter, published paper, manuscript draft, conference paper, research report, literature review, policy brief or technical report.
- Names and contact details of two referees, preferably one academic and one professional or field supervisor.

- Statement of availability for the full eight-month engagement, including availability for ten working days per month and field travel.

15. Suggested Selection Criteria

Criterion	Indicative Weight
Academic fit and discipline relevance	20%
Field research and farmer engagement experience	20%
Data collection, data cleaning and analysis skills	20%
Scientific writing, literature review and publication potential	25%
Availability for 10 days/month, professionalism, ethics and field readiness	15%

16. Equal Opportunity and Safeguarding

Mzuri Organics encourages applications from qualified candidates who demonstrate integrity, respect for farmers, commitment to ethical research and willingness to work in inclusive field settings. The project promotes meaningful participation of women and youth and expects all team members to uphold non-discrimination, safeguarding, confidentiality and do-no-harm principles.

17. Administrative Notes

- This ToR should be read together with the call for applications, the Graduate Assistant contract, project SOPs, consent protocols, data protection rules and field safety instructions.
- Mzuri Organics reserves the right to adjust field schedules, counties, deliverables and reporting templates according to project needs, donor requirements and field realities.
- Final engagement will be subject to successful selection, reference checks where applicable, contract signing and orientation on project ethics, data procedures, publication rules and field protocols.